the Ranger arrive with larger and larger time separations as measured by Brand. After a few hours Brand receives the last bit that she will ever receive from Cooper, the last bit that Cooper emitted before piercing the horizon.

Cooper, by contrast, continues receiving signals from Brand even after he crosses the horizon. Brand's signals have no trouble entering Gargantua and reaching Cooper. Cooper's signals can't get out to Brand. Einstein's laws are unequivocal. This is how it must be.

Moreover, those laws tell us that Cooper sees nothing special as he crosses the horizon. He can't know, at least not with any ease, which bit that he transmits is the last one Brand will receive. He can't tell, by looking around himself, precisely where the horizon is. The horizon is no more distinguishable to him than the Earth's equator is to you as you cross it in a ship.

These seemingly contradictory observations by Brand and Cooper are a result of two things: The warping of time, and the finite travel time for the light and information that they send to each other. When I think carefully about *both* of these things, I don't see any contradiction at all.

Sandwiched Between Singularities



As the Ranger carries Cooper deeper and deeper into the bowels of Gargantua, he continues to see the universe above himself. Chasing the light that brings him that image is an infalling singularity. The singularity is weak at first, but it grows stronger rapidly, as more and more stuff falls into Gargantua and piles up in a thin sheet (Chapter 27). Einstein's laws dictate this.

Below the Ranger is an outflying singularity, created by stuff that fell into the black hole long ago and was backscattered upward toward